Differential Amplifier Design (2025)

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***Abstract*— This experiment focused on the design, simulation, and experimental verification of a differential amplifier using a CA3046 NPN transistor array and a Widlar current source for biasing. The amplifier was required to meet specific performance criteria, including a single-ended voltage gain between 20 and 30, an input resistance of at least 20 kΩ, and a maximized undistorted output swing within the limitations of the circuit's topology and biasing conditions. The amplifier was powered by a ±10 V dual supply and analyzed through theoretical calculations, Multisim simulation, and physical measurement. Key performance parameters such as differential gain, input resistance, and common-mode rejection ratio (CMRR) were evaluated and compared across all methods. The results showed strong agreement between calculated, simulated, and measured values, with minor deviations attributed to component tolerances. The successful design and validation of the differential amplifier demonstrate a practical understanding of analog circuit principles and the importance of accurate biasing and component selection in achieving desired performance.**

I. INTRODUCTION

Differential amplifiers are fundamental building blocks in analog circuit design, offering high common-mode rejection and superior noise immunity compared to single-transistor amplifiers. This experiment focuses on designing and analyzing a differential amplifier that meets specific performance constraints: a single-ended voltage gain in the range of $20 \leq A_d \leq 30$, an input resistance $R_{id} \geq 20$ kΩ, and a maximum undistorted output swing constrained by the circuit topology and biasing conditions. The amplifier operates with a ± 10 V dual power supply and employs a Widlar current source for biasing to ensure stable operation.

The primary objective is to design the amplifier to meet these constraints, simulate its behavior in Multisim, and experimentally validate its performance. Key performance metrics include the differential voltage gain, input resistance, common-mode gain, and common-mode rejection ratio (CMRR). These parameters are critical in applications where differential signal processing is required, such as sensor interfacing and analog signal conditioning. Experimental measurements are compared with theoretical calculations and simulation results to analyze deviations and assess the accuracy of the design.

II. SYSTEM DESCRIPTION

The differential amplifier consists of a matched NPN transistor pair from the CA3046 transistor array, configured in a common-emitter differential pair topology. A Widlar current source biases the emitter terminals, providing a stable and low-current bias while improving thermal stability. The input signals are applied to the transistor bases, and the differential output is taken from the collector of one transistor. A resistive load at the collector converts the amplified current signal into a voltage output. The circuit operates with a ± 10 V dual power supply, ensuring sufficient headroom for signal swing. A common-mode voltage test is performed by applying identical signals to both inputs, allowing for the measurement of common-mode gain and the calculation of the common-mode rejection ratio (CMRR). The full circuit seen in Figure 1 has it's performance evaluated through simulation in MultiSim.

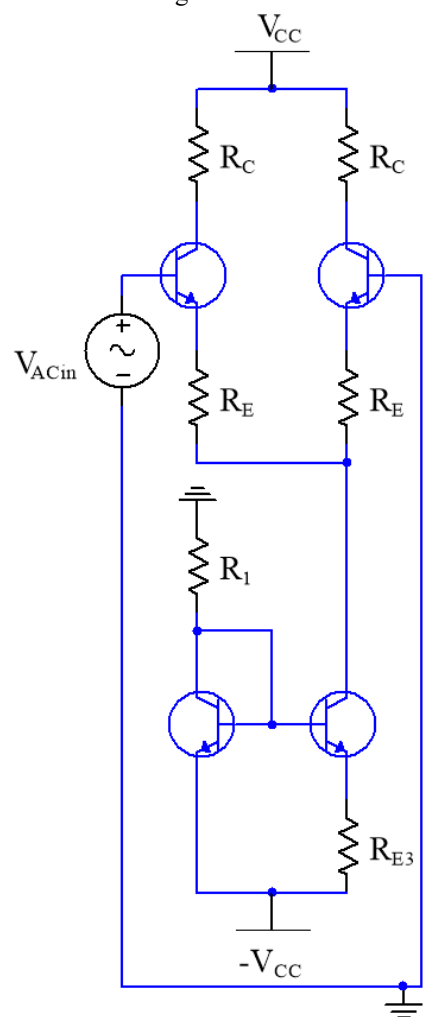

Figure 1: Designed Circuit

A. Circuit Components Calculation

Assuming β = 100. (1) shows the equation for R_{id}.

$$R_{id} = 2(\beta + 1)(R_E + r_e) = 2(\beta + 1)\left(R_E + \frac{V_T}{I_C\left(\frac{\beta}{\beta+1}\right)}\right) \quad (1)$$

This leads to the result in (2):

$$R_{id} = 2(100+1)\left(R_E + \frac{0.026}{I_C\left(\frac{100}{100+1}\right)}\right) = 202R_E + \frac{5.3045}{I_C} \quad (2)$$

Given in the lab requirements, (3) must be true, which can be then utilized to find (

$$R_{id} \geq 20\text{k}\Omega \quad (3)$$

$$202R_E + \frac{5.3045}{I_C} \geq 20\text{k}\Omega \quad (4)$$

The voltage gain of the single ended output is seen in (4). Parameters were given as shown in (5):

$$20 \leq A_d \leq 30 \quad (5)$$

$$40 \leq A_v \leq 60 \quad (5)$$

From this a range of (6)-(7) can be determined:

$$40 \leq \frac{R_C}{\frac{(100+1)0.026}{100I_C} + R_E\left(\frac{100+1}{100}\right)} \leq 60 \quad (6)$$

$$40 \leq \frac{R_C}{\frac{0.02626}{I_C} + R_E(1.01)} \leq 60 \quad (7)$$

Based on this, (8) must also be true.

$$\frac{R_C}{R_E(1.01)} \leq 60 => \frac{R_C}{R_E} \leq 60.6 \quad (8)$$

For max swing, an I_C in the middle of its range is ideal. This means that ideally the voltage across R_C is equal to the voltage across the transistor, and this voltage is maximized. This can be determined by utilizing (9) assuming minimal voltage across R_E and a $V_{CE(sat)}$ of 0.2 V:

$$V_{R_C} \lesssim \frac{V_{CC} - V_{CE(sat)}}{2} = \frac{10-0.2}{2} = 4.9\ V \quad (9)$$

To ensure the transistor does not enter saturation, a V_{RC} of 4.75 V is chosen. This gives (10):

$$I_C = \frac{V_{R_C}}{R_C} = \frac{4.75}{R_C} => I_C R_C = V_C = V_{CE} = 4.75 \quad (10)$$

Utilizing these calculated ratios in (8) and (10), (11) can be determined, later derived to be (12):

$$202R_E + \frac{5.3045}{\frac{4.75}{60.6R_E}} \geq 20\text{k}\Omega \quad (11)$$

$$202R_E + 10.7368R_E => R_E \geq 74.143\Omega \quad (12)$$

From this an $R_E = 100$ can be chosen to ensure the R_{id} stays above 20 kΩ. This can then be utilized with the ratios in (9) and (12), resulting in (13) and (14):

$$\frac{R_C}{100} \leq 60.6 => R_C \leq 6{,}060 \quad (13)$$

$$I_C 5{,}600 = 4.75 => I_C = 0.84\ \text{mA} \quad (14)$$

From (14), an $R_C = 5{,}600\ \Omega$ as it is a value in the lab. A value as close to the max R_C was chosen to maximize the voltage gain. This was then used in (15).

To design the circuit first the reference current (I_{REF}) was established. An $I_{REF} = 10$mA was chosen to ensure accuracy and reduce the sensitivity to varying impedances, the reference current should be much greater than the resulting current of 1mA. β/β+1 is assumed to be insignificant in all calculations

$$10I_o \leq I_{REF} \quad (1)$$

A $V_{CC} = 10$V is used. Utilizing (2) for the greatest symmetrical swing, and the constraint of an $I_o = 1$mA, R_E can be calculated, as seen in (3).

$$I_o R_{E3} = V_T ln\left(\frac{I_{REF}}{I_o}\right) \quad (2)$$

$$R_{E3} = \frac{0.026ln\left(\frac{0.010}{0.00168}\right)}{0.000168} = 27.60629 \quad (3)$$

A value of 27 Ω will be used as it is in the lab. Assuming a $V_{BE} = 0.7$V, utilizing (4), the R_1 value is derived as seen in (5).

$$R_1 = \frac{V_{CC} - V_{BE}}{I_{REF}} \quad (4)$$

$$R_1 = \frac{10-0.7}{0.010} = 930 \quad (5)$$

A value of 910 Ω will be used as it is in the lab.

Figure 2 shows R_{id} over different frequencies. Figure 3 shows the half voltage gain of the circuit. Figure 4 shows the V_o at different inputs. Figure 5 shows the V_{CE} and the I_C of the circuit.

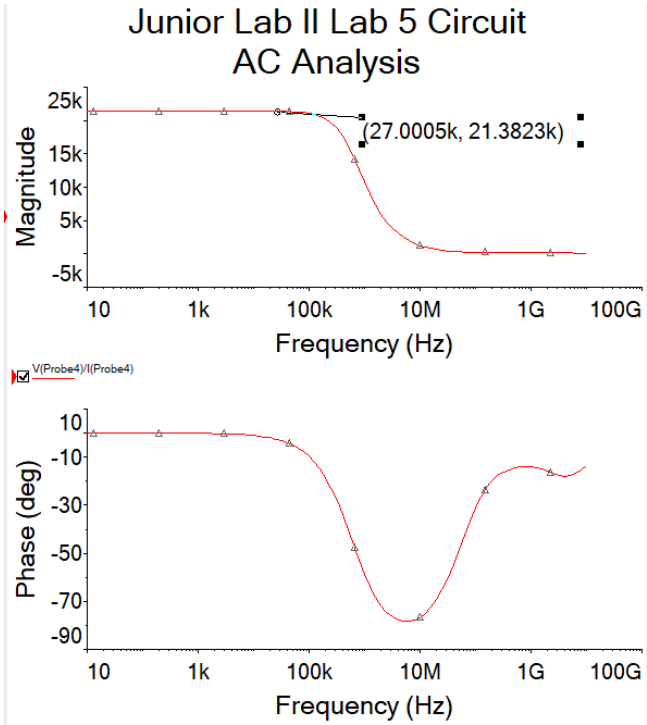

Figure 2: R_{id} at different frequencies

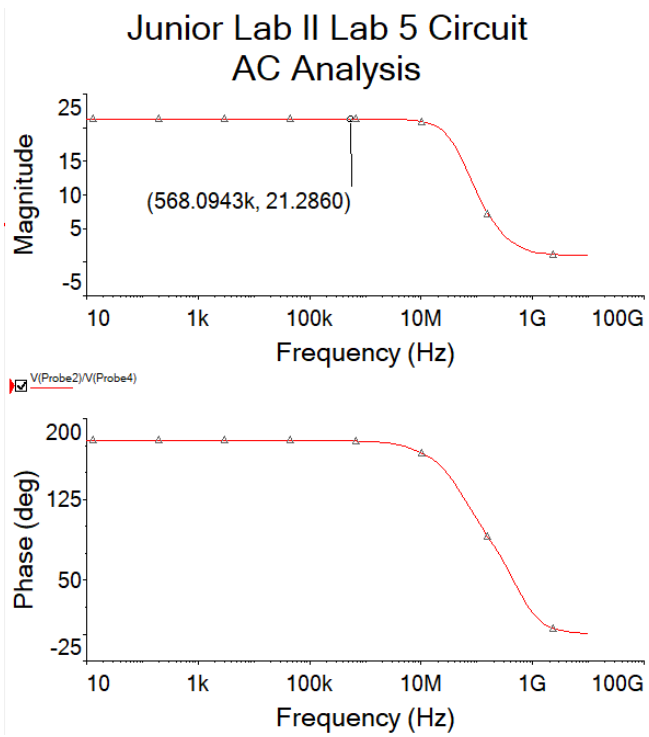

Figure 3: Half Voltage gain

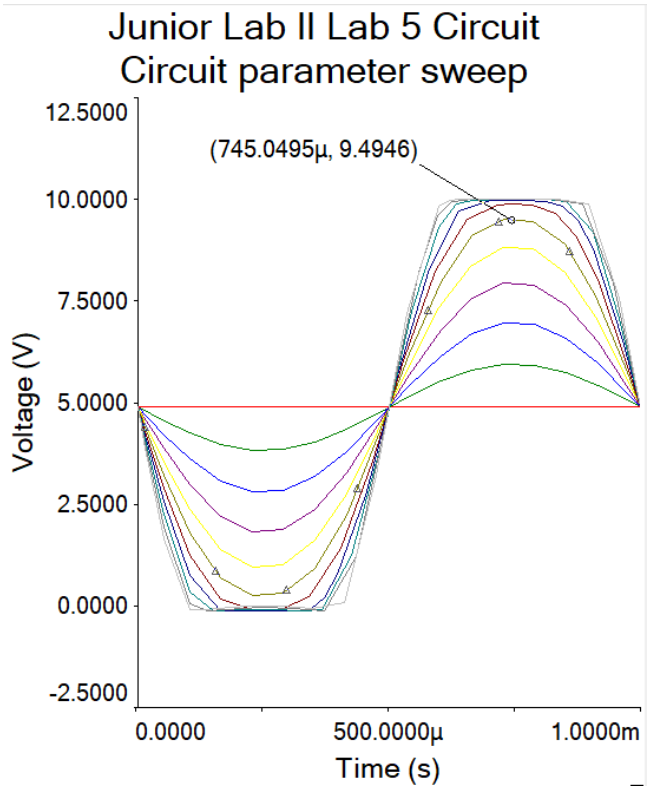

Figure 4: V_o at different inputs (point is with an input of 0.25mV)

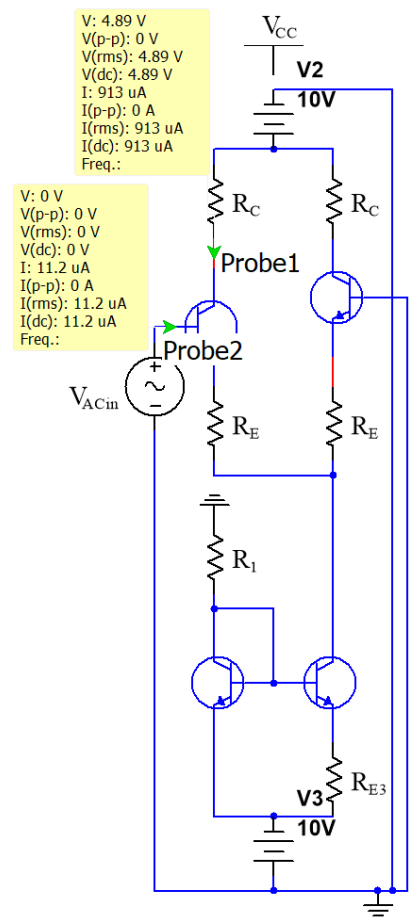

Figure 5: V_{CE} and I_C of the circuit

B. Verification

These equations were used to verify the circuit.

$$R_{id} = 202R_E + \frac{5.3045}{I_C} = 202(100) + \frac{5.3045}{0.000168} = 23{,}357 \quad (1)$$

$$40 \leq \frac{5{,}600}{\frac{(100+1)0.026}{100(0.00084)} + 100\left(\frac{100+1}{100}\right)} \leq 60 \quad (2)$$

$$20 \leq 21.17 \leq 30 \quad (3)$$

$$V_i = \frac{V_o}{A_d} = \frac{2(5.296)}{42.34} = 0.25mV \quad (4)$$

$$Max\ output\ swing = I_C R_C = 0.84(5.6) = 4.704V \quad (5)$$

III. Experimental Evaluation

A. List of Materials

-CA3046 NPN Transistor Chip
-DC Power Supply
-Signal Generator
-Standard Resistors & Capacitors
-Breadboard

B. Experimental Procedure

-Build the current source and verify its proper operation
-Measure the laod current for the limiting load resistances of the circuit
-Build the complete differential amplifier
-Measure single ended input voltage and single ended output voltage
-Measure differential input resistance Rid
-Measure the single ended and double ended voltage gains

C. Results

The following table displays the simulated, calculated, and measured values for the components, bias values, and specifications of the differential amplifier.

Table 1: Simulated, Calculated, & Measured Results

Variable	Simulated	Calculated	Measured
R_{ref}	910 Ω	910 Ω	906.44 Ω
R_{E1}	100 Ω	100 Ω	100.150 Ω
R_{E2}	100 Ω	100 Ω	100.087 Ω
R_{E3}	27 Ω	27 Ω	27.178 Ω
R_{C1}	5.6 kΩ	5.6 kΩ	5.344 kΩ
R_{C2}	5.6 kΩ	5.6 kΩ	5.5139 kΩ
V_{ref}	9.25 V	9.26 V	9.221 V
V_{C1}	5.11 V	5 V	5.45 V
V_{C2}	5.11 V	5 V	5.48 V
V_{E1}	99.8 mV	100 mV	100.4 mV
V_{E2}	99.8 mV	100 mV	98.9 mV
V_{E3}	51.9 mV	52 mV	54.3 mV
I_{ref}	10.1 mA	10 mA	10.173 mA
I_{C1}	0.913 mA	0.84 mA	0.9848 mA
I_{C2}	0.913 mA	0.84 mA	0.9939 mA
I_{EE}	1.72 mA	1.68 mA	1.9786 mA
R_{id}	23.623 kΩ	23.357 kΩ	29.0916 kΩ
Max Swing	9.4946	9.408	9.164
A_d CMSE	0.180	0.186	0.224
A_d CMDE	0.181	0.188	0.225
A_d DMSE	20.6	21.17	20.65
A_d DMSE	20.6	21.17	20.99
CMRR (Single)	114.44	113.82	92.08
CMRR (Double)	113.81	112.61	93.20

D. Discussion of Results

As can be seen in the attachments in Figure 6, the max swing without distortion was found to be 9.164 V peak to peak. The input and output AC voltages utilized to measure the voltage gains can all be seen in the attachments Figure 7 through Figure 14. The single ended differential voltage gain is a little higher than expected, as the calculated and simulated values are approximately 0.4 less than the measured results. The measured input resistance is higher than expected, likely due to the innaccurate component values. Overall, the measured values aligned with the expected simulated & calculated values. The minor differences can be attributed to the component values not being exactly the same as the nominal values. The common mode rejection ratio (CMRR) for the single ended output is 92.08, and 93.20 for the double ended output. The CMRR is therefore better for the double ended output, as a higher CMRR indicates better noise rejection.

E. Limitations

As stated prior, the component values do not exactly align with the nominal values, this contributes to the differential amplifier's performance not being exactly as expected. The differential amplifier characteristics make noise rejection stronger compared to a common emitter amplifier for example. This correlates to the measured gain values shown in Table 1, as the noise can significantly impact the gain values which in turn affects the CMRR, which reflects how well a given amplifier handles noise rejection.

IV. Conclusion

The design, simulation, and experimental evaluation of the differential amplifier were successfully completed, meeting the primary objectives of the lab. Through careful selection of component values and biasing strategies, the amplifier achieved a single-ended voltage gain within the target range of 20–30, maintained an adequate input resistance, and maximized the undistorted output swing while operating under a ±10 V dual power supply. The use of a Widlar current source provided stable biasing, which contributed to consistent performance

across theoretical, simulated, and measured results. The measured common-mode rejection ratio (CMRR) demonstrated strong noise rejection capabilities, with slightly higher performance observed in the double-ended configuration. Minor discrepancies between calculated, simulated, and experimental values were primarily attributed to component tolerances and real-world non-idealities.

References

[1] N. Ula, Introduction to ARM Assembly Language programming using Raspberry Pi, Loyola Marymount University, EECE3200-Lab01Tutorial (1).pdf, 2025

[2] OpenAI, ChatGPT, Loyola Marymount University, ChatGPT, 2025

Attachments

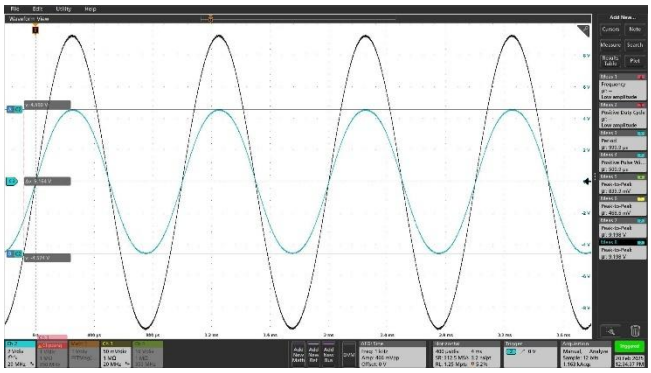

Figure 6: Max output before distortion

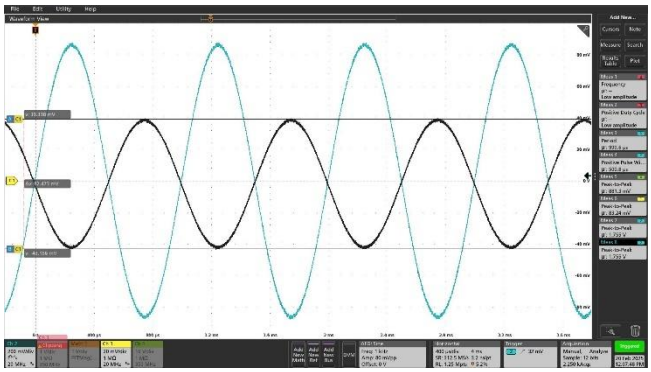

Figure 7: V_{in} for Differential mode double ended voltage gain

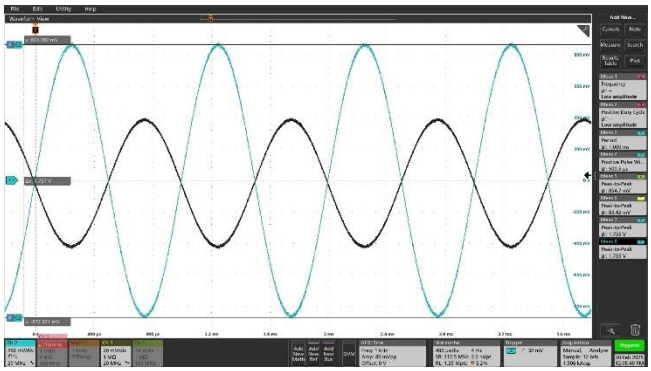

Figure 8: V_{out} for Differential mode double ended voltage gain

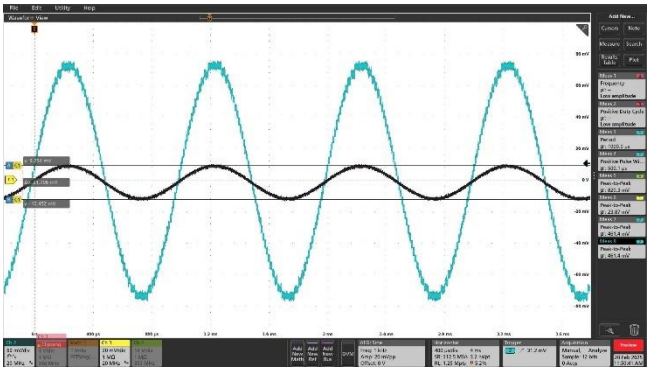

Figure 9: V_{in} for Differential mode single ended voltage gain

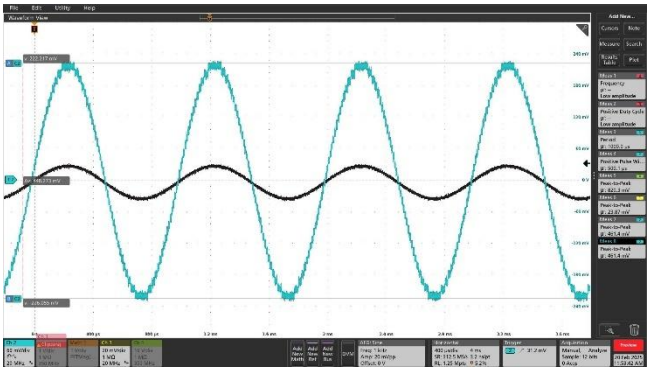

Figure 10: V_{out} for Differential mode single ended voltage gain

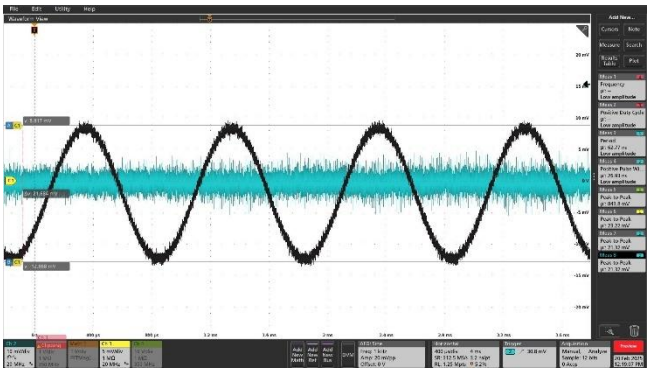

Figure 11: V_{in} for Common mode double ended voltage gain

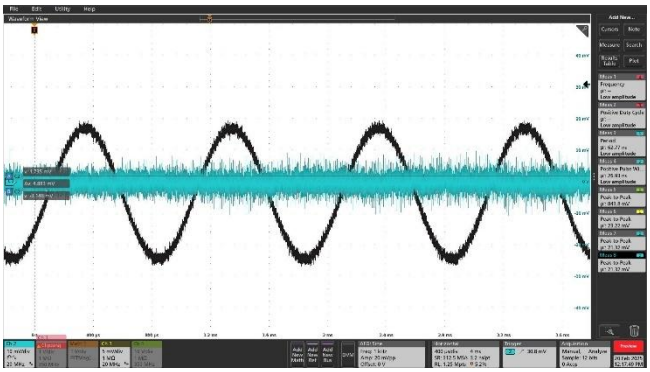

Figure 12: V_{out} for Common mode double ended voltage gain

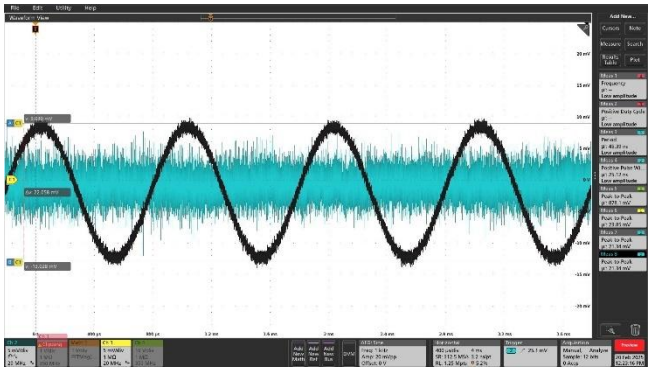

Figure 13: V_{in} for Common mode single ended voltage gain

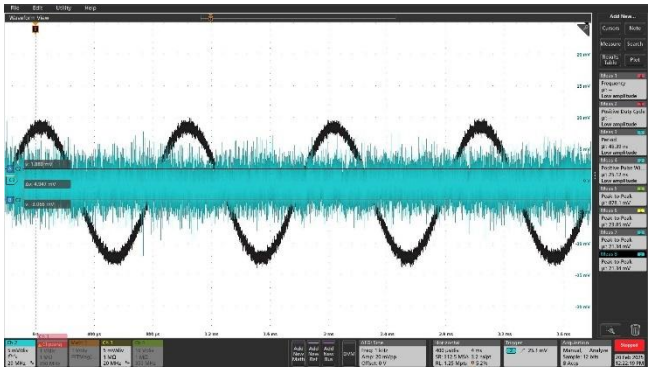

Figure 14: V_{out} for Common mode single ended voltage gain